

THE PILLOW FIELDS · EMPEROR BEAR · EVERY UNIT, EVERY DIFFICULTY

What every Snugg, cub, paratrooper, guard, Bluecoat, cannon, cavalryman and Emperor Bear does to your men on EASY, MEDIUM, HARD and MEGA, with every body's HP and head-count · read out of the live game v28.88 on Nov 1, 2026 03:48 EST · every damage number checked on the game's own machines (Chrome 329/329 · WebKit 329/329) · for Raziya

The one thing to know: in the Pillow Fields **every blow of the bears' bodies is a KO, whatever a man's HP**: the cub's pounce, the paratrooper's landing, a Bluecoat's sword, the hooves of the Emperor and his cavalry, and the Emperor's own sword (three cuts a lunge, three men down). **Sarge is no exception here**: under a moving horse any Sarge and Jane fall, and the Emperor's cut fells a saber Sarge without a block roll. Only three hits take part of a man: a Snugg's or a paratrooper's **glob (2 bar hits**, a man mends one for every 4 s left alone), a cotton cannon's lob (**half his bar**, so three lobs 3.5 s apart), and a Bluecoat's cut on Sarge or a hero (**1 bar hit**). Difficulty changes the bodies (×1 · 1.35 · 1.5 · 1.8), the head-counts (×1 · 1.25 · 1.5 · 1.5) and the lab's level; a hit is the same on every tier. MEGA brings nothing new to a hit: the +20 % and the double weather are the Ice Castle's law, not the fields'. The Emperor fights on **two bars** (152 · 233 · 287 · 390 each, solo at the lab's levels; the same price of the rung rides his Bluecoats, officer and cannons, and **a party pays the tier alone**: 289 · 390 · 433 · 520 with two commanders): one on the horse, one on foot.

THE BOSS LAB fights the Emperor with its stock army at **LEVEL 50 on EASY, 57 on MEDIUM, 63 on HARD and 70 on MEGA**: Sarge with the saber, the jeep, the tank, the dogfighter and the bomber, and seven men at that rank (rifleman, M. gunner, bazooka, mortar, medic, flame, sweeper). The radio man, the grenadier, the commando and the spy are a tap away in THE PLATOON. Every "level" number on this page is that army at that tier's level.

What changed recently (since Oct 29)

live since	when	card	what it does in this fight
v28.74	Oct 31 00:11	**CARD-HP3 · THE HEAVIES**	+3 HP every tenth rank for the bazooka, grenadier, flame and commando. At level 50: bazooka 49, grenadier 45, flame 61, commando 58 HP. A glob or a lob lands a hair lighter on them (the share); the KO strikes do not care.
v28.68	Oct 30 23:31	**CARD-BRB · THE BOMBER**	Your bomber flies in every boss fight now, the Pillow Fields included, and aims at the thickest knot of plush (or the Emperor).
v28.65	Oct 30 19:39	**CARD-LRB · ONE HP NUMBER**	Your man's HP is the number on his unit sheet. A glob, a lob and a Bluecoat's cut on Sarge still count in bar hits, but land a little lighter on tough men: 90 % + 10 % × bar ÷ HP (a level-50 flame takes 1.82 of a glob's 2, a rifleman 1.85).

live since	when	card	what it does in this fight
v28.46	Oct 30 12:57	**CARD-BLP · THE BOSS LAB'S STOCK ARMY**	The lab fights the Emperor with its own army at LEVEL 50 · 57 · 63 · 70 (the lab's level sheet for the fields reads 50–70, spread over the four tiers): the one this page reads. Past rung 50 that level also prices his bars.
read	Oct 10	**CARD-BHP · ONE BOSS, MORE HEALTH**	A party does not multiply the Emperor: it grows his bars (×1 + 0.9 a commander past the first); the cavalry's bar grows the same way (the table below).
—	—	**UNCHANGED SINCE THEN**	The plush's hits, the Bluecoats' blade, the cotton cannon's half-bar lob (CARD 0902, Sep 2), the hooves that fell Sarge and Jane (0829 / 0904, Sep 3–5), the hunt on foot (0904), the royal funnel and the battery (0902), the wave sheets. The stone wall and the reflect are switched off (0904). This Mac has no git log for the game: the dates above come from the other breakdowns and the client's own comments.

How a hit is counted

Your men carry **HP**: the number on their unit sheet (a level-50 flame has 61, a rifleman 26; at level 70: 80 and 33). Their health bar is also cut into a few **bar hits** (7 for most men at these levels, 6 for the medic and spy, 8 for the commando, 12 for the saber Sarge). Three kinds of hit in this fight:

- **KO hits** (the cub's pounce, the paratrooper's landing, a Bluecoat's cut on a man, the hooves, the Emperor's sword and saddle swing): out in one hit whatever his HP.
- **Bar hits** (a glob: 2; a Bluecoat's cut on Sarge or a hero: 1; the Emperor's or the cavalry's blow on a hero: 1): since v28.65 a bar hit lands 90 % + 10 % × bar ÷ HP, so a level-50 rifleman takes 1.85 of his 7 from a glob (6.9 HP of 26). A man who takes no hit for 4 s mends one bar hit, and again every 4 s.
- **The cotton lob**: half the man's bar × that share, so 3.24 of a level-50 rifleman's 7: three lobs 3.5 s apart KO every man (two KO a hero, whose share is 1); the jeep and the tank take one hull hit a lob.

Saves still roll on every hit: a jetting, dropping or dragged man cannot be touched, the pet shield, the deflect dome, the absorbs, and **the saber Sarge's block turns a glob, a lob, a pounce or a Bluecoat's cut 50 % of the time** (measured: 27, 19, 23, 16 of 40 pounces turned on the four tiers on Chrome, 14, 19, 18, 28 on WebKit: 14–28 of 40, a coin's spread about 20). **It never turns the Emperor's cut or a moving horse**: those come already rolled (measured 10 KO of 10 cuts, 0 blocks; every Sarge and Jane under the hooves on every tier). The Claymore Sarge's counted block and the riot plate keep their say. A Bluecoat whose man is a saber Sarge mid-charge waits for the strike to land, then cuts.

The strikes · what a hit does by difficulty

hit	EASY	MEDIUM	HARD	MEGA	who it hits	since
THE CUB'S POUNCE	**KO**	**KO**	**KO**	**KO**	the man he hops on, then the next within 150 px	unchanged
THE SNUGG'S GLOB · bar hits	**2 × share**	**2 × share**	**2 × share**	**2 × share**	one man within 280 px, every 3.6 s	v28.65: lands 90 % + 10 % × bar ÷ HP of that
THE PARATROOPER'S THUMP	**KO**	**KO**	**KO**	**KO**	every man within 76 px of his landing (the shadow sways up to 9 px)	unchanged
THE PARATROOPER'S GLOB · bar hits	**2 × share**	**2 × share**	**2 × share**	**2 × share**	one man within 230 px, every 2.6 s	unchanged
A BLUECOAT'S CUT	**KO**	**KO**	**KO**	**KO**	the nearest man within 74 px of each cut, three cuts a lunge	unchanged
A BLUECOAT'S CUT · Sarge, a hero	**1 × share**	**1 × share**	**1 × share**	**1 × share**	a bar hit through his saves (the saber turns half)	unchanged
THE COTTON LOB · bar hits	**½ bar × share**	**½ bar × share**	**½ bar × share**	**½ bar × share**	every man within 50 px of the burst; the jeep: a hull hit	unchanged
THE HOOVES (Emperor · cavalry)	**KO**	**KO**	**KO**	**KO**	a man within 52 px of a moving horse; Sarge and Jane fall; a hero 1 bar hit + the throw a pass (the Emperor's hooves land it every pass, the cavalry never a hero's last)	unchanged
THE EMPEROR'S SADDLE SWING (at rest)	**KO**	**KO**	**KO**	**KO**	a man within 52 px every 0.6 s; a Sarge with a blade in hand: 1 bar hit and keeps his feet; a gun Sarge, a hero: 1 bar hit + thrown 135–185 px (the pistol Sarge 181–185, King Rex 135; measured)	unchanged
THE EMPEROR'S SWORD	**KO**	**KO**	**KO**	**KO**	three cuts a lunge, each the nearest uncut man within 74 px; Sarge: the block never rolls	unchanged
BODY HP · the plush	**×1**	**×1.35**	**×1.5**	**×1.8**	the guards, Snuggs, paratroopers, cubs and cavalry: the tier alone	unchanged
BODY HP · the Emperor, his Bluecoats, officer and cannons	**×1 × 1**	**×1.35 × 1.136**	**×1.5 × 1.257**	**×1.8 × 1.422**	the tier × THE PRICE of the player's rung past 50 (×1 · 1.136 · 1.257 · 1.422 solo at the lab's levels 50 · 57 · 63 · 70; ×1 in a party or at rung 50 or under); the Emperor's bars at full weight	unchanged

hit	EASY	MEDIUM	HARD	MEGA	who it hits	since
HEAD-COUNTS	**×1**	**×1.25**	**×1.5**	**×1.5**	the Snuggs and the paratroopers; the guards, the cubs and the cavalry stay 4 · 3 · 3	unchanged
THE EMPEROR'S BAR (lab's level, one commander)	**152 × 2 lives**	**233 × 2 lives**	**287 × 2 lives**	**390 × 2 lives**	152 × the tier × the price of the player's rung (×1 · 1.136 · 1.257 · 1.422 at level 50 · 57 · 63 · 70)	unchanged

share = 90 % + 10 % × the man's bar hits ÷ his HP (CARD-LRB). The fields' dials sit in THE BOSS LAB's RULES under THE PILLOW FIELDS: a typed cell is what the next lab fight uses.

Who survives · the lab's levels 50 · 57 · 63 · 70

How many of that hit in a row KO a fresh man at the tier's own level. **KO** = one hit. Hover a cell for the numbers.

KO one hit · 2 two · 3 three · 4 four · 5+ five or more · a dot in the corner = measured on the game's own machine in this read · ½ = the saber turns half

	HP / BAR HITS AT THE LAB'S LEVEL				GLOBS TO KO (Snugg, paratrooper)				CANNON LOBS TO KO						
man	EASY 50	MEDI 57	HARD 63	MEGA 70	E	M	H	G	E	M	H	G	BLUECOAT'S CUT	POUNCE · THUMP	HOOVES · SWORD
THE LAB'S STOCK ARMY															
RIFLEMAN	26 / 7	26 / 7	29 / 7	33 / 7	4	4	4	4	3	3	3	3	KO	KO	KO
M. GUNNER	40 / 7	40 / 7	45 / 7	50 / 7	4	4	4	4	3	3	3	3	KO	KO	KO
BAZOOKA	49 / 7	49 / 7	58 / 7	66 / 7	4	4	4	4	3	3	3	3	KO	KO	KO
MORTAR	23 / 7	23 / 7	26 / 7	30 / 7	4	4	4	4	3	3	3	3	KO	KO	KO
MEDIC	22 / 6	22 / 6	26 / 6	29 / 6	4	4	4	4	3	3	3	3	KO	KO	KO
FLAME	61 / 7	61 / 7	70 / 7	80 / 7	4	4	4	4	3	3	3	3	KO	KO	KO
SWEEPER	24 / 7	24 / 7	28 / 7	31 / 7	4	4	4	4	3	3	3	3	KO	KO	KO
SARGE · SABER	35 / 12	35 / 12	37 / 12	40 / 12	7	7	7	7	3	3	3	3	13	KO ½	KO
A TAP AWAY (THE PLATOON, SARGE'S OTHER IRONS)															
RADIO MAN	24 / 7	24 / 7	28 / 7	31 / 7	4	4	4	4	3	3	3	3	KO	KO	KO
GRENADIER	45 / 7	45 / 7	53 / 7	61 / 7	4	4	4	4	3	3	3	3	KO	KO	KO
COMMANDO	58 / 8	58 / 8	66 / 8	75 / 8	5	5	5	5	3	3	3	3	KO	KO	KO

	HP / BAR HITS AT THE LAB'S LEVEL				GLOBS TO KO (Snugg, paratrooper)				CANNON LOBS TO KO						
man	EASY 50	MEDI 57	HARD 63	MEGA 70	E	M	H	G	E	M	H	G	BLUECOAT'S CUT	POUNCE · THUMP	HOOVES · SWORD
SPY	**27** / 6	**27** / 6	**32** / 6	**36** / 6	4	4	4	4	3	3	3	3	KO	KO	KO
SARGE · PISTOL	**35** / 11	**35** / 11	**38** / 11	**41** / 11	6	6	6	6	3	3	3	3	12	KO	KO
SARGE · SHOTGUN	**35** / 8	**35** / 8	**37** / 8	**40** / 8	5	5	5	5	3	3	3	3	9	KO	KO
SARGE · SNIPER	**26** / 8	**26** / 8	**28** / 8	**31** / 8	5	5	5	5	3	3	3	3	9	KO	KO
SARGE · CLAYMORE	**68** / 23	**70** / 23	**71** / 23	**71** / 23	13	13	13	13	3	3	3	3	25	KO*	KO

GLOBS: by the sum of the hits; a Snugg's globs come 3.6–4.2 s apart and a man mends one bar hit for every 4 s he is left alone, so a lone Snugg often needs one more (measured 4–7 globs on 7-bar men over 40 runs on both engines; 2 runs needed 7, when nearly every gap passed the 4 s mend). LOBS: 3.5 s apart, inside the mend. BLUECOAT'S CUT on Sarge: 1 bar hit through his saves. * The Claymore Sarge's own block can eat a KO whole (not measured here).

The heroes at the lab's level (one in command), measured

hero	HP (E · M · H · G)	bar hits	pounce · thump	a glob	a cotton lob	the cavalry's hooves	the Emperor's sword
KING YETI	**122 · 124 · 131 · 138**	114 · 114 · 120 · 126	KO	2 · 2 · 2 · 2 bar hits	57 · 57 · 60 · 63 of 114 · 114 · 120 · 126 (half)	1 bar hit + thrown 154–160 px a pass	KO
KING REX	**176 · 179 · 200 · 209**	105 · 105 · 131 · 137	KO	2 · 2 · 2 · 2 bar hits	52.5 · 52.5 · 65.5 · 68.5 of 105 · 105 · 131 · 137 (half)	1 bar hit + thrown 158–216 px a pass	KO
KING CROAK	**101 · 103 · 110 · 117**	65 · 65 · 71 · 77	KO	2 · 2 · 2 · 2 bar hits	32.5 · 32.5 · 35.5 · 38.5 of 65 · 65 · 71 · 77 (half)	1 bar hit + thrown 7–27 px a pass	KO
JANE	**108 · 111 · 118 · 126**	56 · 56 · 62 · 68	KO	2 · 2 · 2 · 2 bar hits	28 · 28 · 31 · 34 of 56 · 56 · 62 · 68 (half)	KO (she falls under the hooves)	KO
EMPEROR BEAR (the hero)	**184 · 188 · 209 · 218**	105 · 105 · 131 · 137	KO	2 · 2 · 2 · 2 bar hits	52.5 · 52.5 · 65.5 · 68.5 of 105 · 105 · 131 · 137 (half)	1 bar hit + thrown 95–105 px a pass	KO
PRINCESS CROAK	**82 · 83 · 90 · 97**	65 · 65 · 71 · 77	KO	2 · 2 · 2 · 2 bar hits	32.5 · 32.5 · 35.5 · 38.5 of 65 · 65 · 71 · 77 (half)	1 bar hit + thrown 156–158 px a pass	KO

A hero's share is 1 (his HP is not on the unit sheet), so a glob takes the whole 2 and a lob exactly half his bar: two lobs KO any hero. Jane falls under a moving horse like Sarge; the other heroes take 1 bar hit and are thrown a pass. The cavalry never take a hero's last bar hit; **the Emperor's hooves do**: a hero at his last bar hit falls under his charge (measured KING REX on every tier). The Emperor's sword KOs every hero (measured on every tier).

The level a man first survives

For the hits that take part of a man. The KO hits have no such level: no rank survives them. **Green** = he already does at the lab's EASY level 50; — = not even at rank 100.

			GLOBS IN A ROW (Snugg · paratrooper)				CANNON LOBS IN A ROW			BLUECOAT CUTS IN A ROW (Sarge)		
man	HP / bar at 50	HP / bar at 100	1	2	3	4	1	2	3	4	8	12
RIFLEMAN	26 / 7	45 / 8	0	5	19	100	0	2	—	—	—	—
M. GUNNER	40 / 7	65 / 8	0	5	19	100	0	0	—	—	—	—
BAZOOKA	49 / 7	92 / 8	0	6	19	100	0	2	—	—	—	—
MORTAR	23 / 7	40 / 8	0	6	19	100	0	2	—	—	—	—
MEDIC	22 / 6	40 / 7	3	10	42	—	0	3	—	—	—	—
FLAME	61 / 7	110 / 8	0	5	19	100	0	0	—	—	—	—
SWEEPER	24 / 7	42 / 8	0	6	19	100	0	2	—	—	—	—
RADIO MAN	24 / 7	42 / 8	0	6	19	100	0	2	—	—	—	—
GRENADIER	45 / 7	84 / 8	0	6	19	100	0	2	—	—	—	—
COMMANDO	58 / 8	102 / 9	0	0	10	42	0	0	—	—	—	—
SPY	27 / 6	50 / 7	3	10	42	—	0	3	—	—	—	—
SARGE · SABER	35 / 12	50 / 13	0	0	0	0	0	0	—	0	0	42
SARGE · PISTOL	35 / 11	50 / 12	0	0	10	42	0	0	—	0	42	100
SARGE · SHOTGUN	35 / 8	50 / 9	0	0	10	42	0	0	—	0	42	—
SARGE · SNIPER	26 / 8	40 / 9	0	0	10	42	0	0	—	0	42	—
SARGE · CLAYMORE	68 / 23	78 / 24	0	0	0	0	0	0	—	0	0	0

Read from the game's own HP and bar ladders at every rank 0–100 (the tenth-rank extras included; no gifts, soul or gear), by the sum of the hits with no mend between. Sarge's rows read the iron at the same rank. Three lobs never: half the bar × a share under 1 lands 1.5 bars in three.

The units

unit	attack	what a hit does	HP EASY	MED	HARD	MEGA	n EASY	MED	HARD	MEGA	leak	tags
 **THE HONOR GUARD (bearskins)** walks 40 px/s · 8 HP × the tier	**— (no weapon)** four walk on 1.6 s in and take their seats behind the Emperor (measured: no hit on a rifleman in 5 s)	none: he escorts the Emperor and dives in front of every round meant for him	8	11	12	14	4	4	4	4	2	20
 **THE SNUGG** walks 45 px/s, rushes ×2.2 with nobody in reach	**GLOB** a glob every 3.6 s at a man within 280 px (measured gaps 3.6–4.2 s); a man left alone 4 s mends 1 bar hit	**2 bar hits** × 90 % + 10 % × bar ÷ HP · on a level-50/57/63/70 rifleman 1.85 · 1.85 · 1.85 · 1.84 · the saber Sarge turns half	5	7	8	9	6	8	9	9	1	12
 **THE PARATROOPER** walks 40 px/s, rushes ×2 · cannot be aimed at while he hangs	**THE DROP'S THUMP · GLOB** drops 3 s onto the deep band (x 75–85 % of the mat), tangles 0.6 s, then a glob every 2.6 s within 230 px	**KO** to every man within 76 px of where he lands (60 + 30 % of the man's width; the landing point sways up to 9 px under the chute): measured on eight men per tier and engine, 0 px fell on every run; 50 px fell on every run; 80 px fell on 1 of 8 runs and stood on the rest; 100 px stood on every run; 130 px and beyond stood · then **2 bar hits** a glob	12	16	18	22	2	3	3	3	1	21

unit	attack	what a hit does	HP EASY	MED	HARD	MEGA	n EASY	MED	HARD	MEGA	leak	tags
 **THE CUB** runs 140 px/s · 0.75 HP × the tier (rounds to 1)	**POUNCE** hops 460 px/s at a man within 150 px, strikes at contact, the next man 0.9 s later, no cap	**KO** on every tier (the saber Sarge turns half)	1	1	1	1	6	6	6	6	1	6
 **THE CAVALRY (three, at the keel)** 105 HP × 0.6 × the tier · leaks 6 lives at the box	**THE HOOVES** prances 1.2 s, then charges 420 px/s through the thickest cluster every 2nd leg (rides 180 px/s between); a man within 52 px is trampled, once every 0.9 s each	**KO** to every man he rides over · **any Sarge and Jane fall** under a moving horse · another hero: **1 bar hit** + thrown 95–216 px a pass (King Croak 7–27; measured) · the jeep and the tank ride through untouched	63	85	95	113	3	3	3	3	6	150
 **THE BLUECOAT (the Emperor's funnel)** walks 64 px/s, rushes ×1.5 · 8 HP × the tier × 1.5 × THE PRICE of the rung (the HP columns = solo at the lab's level; a party, or rung 50 or under: 12 · 17 · 18 · 21, measured) · comes in details of 2 (a battalion of 4 + a leader every third), a column behind the officer	**DASH-SLASH** a man within 330 px: a wind-up, a flight at 680 px/s to a pace short, three cuts 0.22 s apart at 74 px; then a 6 s clock	**KO** to a man · **1 bar hit** to Sarge or a hero (the saber turns half) · never a glob: he fights with the blade	12	19	23	30	∞	∞	∞	∞	2	16

unit		attack	what a hit does	HP EASY	HP MED	HP HARD	HP MEGA	n EASY	n MED	n HARD	n MEGA	leak	tags
	**THE BLUECOAT OFFICER** 24 HP × the tier × THE PRICE of the rung (solo at the lab's level; a party: 24 · 32 · 36 · 43, measured) · walks 64 px/s	**DASH-SLASH · THE WHISTLE** one on the field at a time; he rides in with the first detail and every detail that finds none standing; his fall breaks the line	the same cuts as a Bluecoat · his whistle hups every Bluecoat within 300 px and marches them ×1.3 for 4 s	24	36	45	61	∞	∞	∞	∞	2	40
	**THE COTTON CANNON (the Emperor's battery)** rolls 40 px/s west and parks at x 1130 behind the river · 10 HP × the tier × 2.5 × THE PRICE of the rung (solo at the lab's level; a party: 25 · 35 · 38 · 45, measured) · two at the ride's start, more with every wave	**COTTON LOB** a lob every 3 s (+ a 0.5 s wind) at the nearest man within 1100 px, the first 85 % primed; lands within 40 px of him, bursts 50 px wide, flies 0.6–1.4 s	**half the man's bar** × 90 % + 10 % × bar ÷ HP a lob (every man: 3 lobs to KO at the lab's levels; a hero: 2) · the jeep and the tank: one hull hit a lob	25	40	47	64	∞	∞	∞	∞	2	24
	**EMPEROR BEAR** rides 140 px/s, charges 420; walks 190 px/s on foot · leaks 8 lives	**see Emperor Bear below** the hooves and the charge on the horse, the saddle swing at rest; the sword's three cuts on foot; two bars	**KO** every move	152	233	287	390	1	1	1	1	8	250

HP = each body as the fields' own wave groups bear it on that tier (measured in this read: the plush are the tier alone). The Bluecoats, the officer and the cannons come through the Emperor's funnel at ×1.5 (Bluecoat) and ×2.5 (cannon) **× the price of the player's rung, like the Emperor's bar**: the HP columns are the lab's SOLO bodies at level 50 · 57 · 63 · 70 (×1 · 1.136 · 1.257 · 1.422); a party, or a player at

rung 50 or under, meets a Bluecoat of 12 · 17 · 18 · 21, an officer of 24 · 32 · 36 · 43 and a cannon of 25 · 35 · 38 · 45 HP (measured on both engines); at rung 100 solo the price is ×2.167. n = bodies named in the solo wave sheet (one commander); ∞ = as many as the funnel brings while he lives. leak = lives it costs at the toy box. tags = dog tags paid when it falls.

Emperor Bear, move by move

move	EASY	MEDIUM	HARD	MEGA	how it works
THE RIDE	140 px/s	140 px/s	140 px/s	140 px/s	He walks on at the east rim and mounts at once (no court, no speech): the battery of 2 cotton cannons is born a screen and a half in (x 1500), the first pyramid bursts, the horn. Every leg he rides AT a man picked at random east of x 300 (never the toy box), a horse-length short of him, re-aimed as the man moves.
THE HOOVES	**KO**	**KO**	**KO**	**KO**	Every man within 52 px of the moving horse is trampled, once every 0.6 s each: a KO. **Any Sarge (every class) and Jane fall** under a moving horse in the Pillow Fields: the blade's block never rolls. Another hero takes 1 bar hit and is thrown 203–208 px a pass (measured on both engines, every tier; EASY on Chrome: the rifleman KO · the flame KO · the bazooka KO · the medic KO · Sarge KO · Jane KO · King Rex 1 bar hit + thrown 205 px · the machine gunner KO), **and under the Emperor every pass counts**: a hero at his last bar hit falls (measured KING REX at 104 of 105 bar hits: KO under the Emperor's charge on every tier; under a cavalryman's charge he stands untouched, thrown 157 px: the cavalry never take a hero's last). The jeep and the tank ride through untouched.
THE CHARGE	420 px/s	420 px/s	420 px/s	420 px/s	Every 2nd–3rd leg (every 1st–2nd when his bar is priced ×1.5 or more): he rears 0.9 s on the spot (the dust ring, the horn, a red ring on his quarry), then runs the man down at 420 px/s and 120 px past him, the hooves landing on everyone in the way. At 70 % and 35 % of his bar the next leg is a charge now.
THE SADDLE SWING	**KO**	**KO**	**KO**	**KO**	At rest between legs he swings at whoever stands within 52 px: a man is KO'd (measured at 40 px). **A Sarge with a blade in hand** (the saber, the Claymore) takes 1 bar hit and keeps his feet (measured: the saber Sarge, moved 11 px; his block turned the swing on 4 of 8 runs). **A gun Sarge and every hero take 1 bar hit AND the sideways throw** (measured: the pistol Sarge 0.93 bar hit, thrown 185 px; KING REX 1 bar hit, thrown 135 px; on every tier, both engines).
THE ROYAL FUNNEL	every 9–14 s	every 9–14 s	every 9–14 s	every 9–14 s	While he rides, 5 s in and then every 9–14 s a detail of 2 Bluecoats walks on from the east rim on his lane (measured 12 · 19 · 23 · 30 HP each at the lab's four levels: the Bluecoat's body ×1.5 × the price of the rung; a party meets 12 · 17 · 18 · 21), in a column behind THE OFFICER (24 · 36 · 45 · 61 HP; a party 24 · 32 · 36 · 43), who comes with the first detail and whenever none stands. Every third detail is a battalion: 4 Bluecoats behind a flagged leader, and a fresh cannon if his battery is short (measured). The next detail waits while more than 6 Bluecoats stand.
THE KEEL	life 1 → 2	life 1 → 2	life 1 → 2	life 1 → 2	At 0 on the horse he is not felled (**measured through the game's own door on every tier**): the rocker keels 0.8 s (nothing lands on him), EN GARDE! in his window, the horn, the second pyramid bursts, and he rises on foot with a **second full bar**, the same number as the first (152 · 233 · 287 · 390). The EN GARDE! wave walks on at the keel (released 0.4–1.2 s after it, measured): 4 · 5 · 6 · 6 Snuggs, 3 cubs and the 3 cavalrymen (bar 63 each on EASY).
THE HUNT ON FOOT	190 px/s	190 px/s	190 px/s	190 px/s	He runs at a man picked at random east of x 300 (never west of it, never over the water: he crosses at the planks), stops at the sword's reach (74 px) and strikes; a man who falls or leaves is re-picked at once. Measured: a rifleman 150 px off is reached and KO'd in 0.433 s.

move	EASY	MEDIUM	HARD	MEGA	how it works
THE LUNGE · THREE CUTS	**KO** × 3 men	**KO** × 3 men	**KO** × 3 men	**KO** × 3 men	A pick within 330 px on his bank: a 0.05 s wind-up (the sabre back over his shoulder, a steel ring at his feet), a flight at 680 px/s to a pace short of him, then **three cuts 0.22 s apart**, each the nearest man not yet cut within 74 px in front: a KO each (measured: three men in reach, 3 KO'd; a rifleman 300 px off KO'd 0.633 s after the clock was ready). The clock is 4.5 s; once in reach he strikes 0.8 s into it. A saber Sarge in reach is KO'd: **the blade's block never rolls on a king's cut** (measured 10 of 10); every hero the same (measured, all six).
MEN WITHOUT FEAR	every 8–15 s	every 8–15 s	every 8–15 s	every 8–15 s	While he stands, every 8–15 s a wave of 3 Bluecoats (at most 12 on the field) and 2 cotton cannons walks on from the east (the cannons born at x 1660); the next wave waits while more than 4 of the last stand. At 75 % of his second bar a wave comes now. **THE FINAL PHASE** (the second bar at or under half): every 6–10 s and no waiting (measured).
THE GATES	70 · 35 % / 75 · 50 · 25 %	70 · 35 % / 75 · 50 · 25 %	70 · 35 % / 75 · 50 · 25 %	70 · 35 % / 75 · 50 · 25 %	On the horse: at 70 % and 35 % a charge now. On foot: at 75 % a wave now, at 50 % the stone wall (OFF today), at 25 % a lunge now.
THE STONE WALL · THE REFLECT	off	off	off	off	Both switches are off in the client today (the stone wall off, the reflect off): he neither braces against your rounds nor catches rockets and mortar bombs.
HIS BARS (one commander, the lab's level)	**152** twice	**233** twice	**287** twice	**390** twice	152 HP × the tier's ×1 · 1.35 · 1.5 · 1.8 at full weight × the price (below) — two full bars, one after the other. With more commanders both grow (the table below). His fall ends the fight; the Bluecoats and the cannons run for the rim, the plush go home.
THE PRICE OF THE RUNG	×1	×1.136	×1.257	×1.422	Solo, a player past campaign rung 50 fights an Emperor priced like a boss of his own rung (the lab's stock army is level 50 · 57 · 63 · 70, so the lab pays ×1 · 1.136 · 1.257 · 1.422); the ladder: rung 50 ×1 · rung 60 ×1.196 · rung 70 ×1.422 · rung 80 ×1.678 · rung 90 ×1.937 · rung 100 ×2.167, capped ×4. **The same price rides his Bluecoats, his officer and his cannons** (the units table). **A party pays the tier alone** (the table below). At rung 50 or under the bars read 152 · 205 · 228 · 274.
THE TOY BOX	–8 lives if he reaches it	–8 lives if he reaches it	–8 lives if he reaches it	–8 lives if he reaches it	He never marches the box: his legs run at your men. A cavalryman leaks 6, a Bluecoat, an officer, a guard or a cannon 2, a Snugg, a paratrooper or a cub 1.

His bars by the number of commanders (the Emperor, each of his two bars · a cavalryman)

	1 commander	2 commanders	3 commanders	4 commanders	5 commanders	10 commanders
EASY a real party (the tier alone)	152 · cavalry 63	289[m] · cavalry 120	426[m] · cavalry 176	562 · cavalry 233	699[m] · cavalry 290	1,383 · cavalry 574
EASY · THE BOSS LAB's party slider (a solo fight, the price ×1 rides)	152	289	426	562	699	1,383
MEDIUM a real party (the tier alone)	233 · cavalry 85	390[m] · cavalry 161	575[m] · cavalry 238	759 · cavalry 314	944[m] · cavalry 391	1,867 · cavalry 774
MEDIUM · THE BOSS LAB's party slider (a solo fight, the price ×1.136 rides)	233	443	653	862	1,072	2,121
HARD a real party (the tier alone)	287 · cavalry 95	433[m] · cavalry 179	638[m] · cavalry 265	844 · cavalry 350	1,049[m] · cavalry 435	2,075 · cavalry 860
HARD · THE BOSS LAB's party slider (a solo fight, the price ×1.257 rides)	287	544	802	1,061	1,319	2,608
MEGA a real party (the tier alone)	390 · cavalry 113	520[m] · cavalry 215	766[m] · cavalry 317	1,012 · cavalry 419	1,259[m] · cavalry 521	2,490 · cavalry 1,032
MEGA · THE BOSS LAB's party slider (a solo fight, the price ×1.422 rides)	390	739	1,089	1,439	1,790	3,541

A real party pays the tier alone: the price of the rung is ×1 while a party launches (the game's twin-sim law: every commander's sim builds the same army from the tier), so from 2 commanders up each of his two bars is 152 × the tier × (1 + 0.9 a commander past the first), read from the game's own sheet with the party flag set; [m] = the Emperor born in a party of that size was measured at exactly this bar on every tier, both engines. One commander is a solo fight, so his bar carries the price of the lab's level. THE BOSS LAB's own party slider (the second line) is a solo fight with a party's bar, so it keeps the solo price: it does not show what a real party meets past rung 50. Each of his two bars is this number; the cavalry's bar never carries the price.

The cavalry's hooves on a line of eight, measured (DUKE BEAR's brain: the prance, then the charge through the line)

	a cavalryman's bar	who fell
EASY	63	the rifleman KO · the flame KO · the bazooka KO · the medic KO · the machine gunner KO · Sarge KO · Jane KO · King Rex 1 bar hit + thrown 216 px
MEDIUM	85	the rifleman KO · the flame KO · the bazooka KO · the medic KO · the machine gunner KO · Sarge KO · Jane KO · King Rex 1 bar hit + thrown 216 px
HARD	95	the rifleman KO · the flame KO · the bazooka KO · the medic KO · the machine gunner KO · Sarge KO · Jane KO · King Rex 1 bar hit + thrown 215 px
MEGA	113	the rifleman KO · the flame KO · the bazooka KO · the medic KO · the machine gunner KO · Sarge KO · Jane KO · King Rex 1 bar hit + thrown 216 px

The funnel, measured

	what walked on
EASY	battery 2 × 25 HP at x 1500 · first detail 2 Bluecoats 12 HP + the officer 24 HP · the battalion 5 + 1 cannon · on foot: 3 Bluecoats + 2 cannons (25 HP), the final phase next in 7.52 s
MEDIUM	battery 2 × 40 HP at x 1500 · first detail 2 Bluecoats 19 HP + the officer 36 HP · the battalion 5 + 1 cannon · on foot: 3 Bluecoats + 2 cannons (40 HP), the final phase next in 6.07 s
HARD	battery 2 × 47 HP at x 1500 · first detail 2 Bluecoats 23 HP + the officer 45 HP · the battalion 5 + 1 cannon · on foot: 3 Bluecoats + 2 cannons (47 HP), the final phase next in 8.94 s
MEGA	battery 2 × 64 HP at x 1500 · first detail 2 Bluecoats 30 HP + the officer 61 HP · the battalion 5 + 1 cannon · on foot: 3 Bluecoats + 2 cannons (64 HP), the final phase next in 8.14 s

What changes from EASY to MEGA

	EASY	MEDIUM	HARD	MEGA
Every hit (a KO is a KO; a glob 2 bar hits; a lob half the bar)			the same on every tier	
Body HP (the plush: the tier alone)	×1	×1.35	×1.5	×1.8
The price of the rung on the Emperor, his Bluecoats, officer and cannons (solo at the lab's level; ×1 in a party)	×1	×1.136	×1.257	×1.422
The Emperor's two bars at the lab's level (solo)	152	233	287	390
The Emperor's two bars with 2 commanders (a real party: the tier alone)	289	390	433	520
A Bluecoat · the officer · a cannon, solo at the lab's level	12 · 24 · 25	19 · 36 · 40	23 · 45 · 47	30 · 61 · 64
… in a party, or at rung 50 or under (measured)	12 · 24 · 25	17 · 32 · 35	18 · 36 · 38	21 · 43 · 45
A cavalryman's bar	63	85	95	113
Snuggs (the opening + EN GARDE!)	2 + 4	3 + 5	3 + 6	3 + 6
Paratroopers	2	3	3	3
Guards · cubs · cavalry			4 · 3 + 3 · 3	
Bodies in the sheet, solo (the Emperor and the cavalry not counted)	18	21	22	22

	EASY	MEDIUM	HARD	MEGA
The lab's stock army	level 50	level 57	level 63	level 70
The Emperor's price at that level (solo)	×1	×1.136	×1.257	×1.422
MEGA's door	—	—	—	the whole 50-rung Bear Park ladder climbed
The castle's +20 % per hit and double weather	not here: that is the Ice Castle's own law			

The wave sheets (solo, one commander)

EASY · 18 bodies + the funnel

1. **EMPEROR BEAR TAKES THE FIELD** (ON THE CLOCK)
   4 honor guards, 2 Snuggs, 2 paratroopers, 3 cubs · then EMPEROR BEAR (bar 152 × 2) at 3 s
2. **EN GARDE!** (HOLDS WHILE THE EMPEROR IS ON HIS HORSE)
   4 Snuggs, 3 cubs, 3 CAVALRYMEN

MEDIUM · 21 bodies + the funnel

1. **EMPEROR BEAR TAKES THE FIELD** (ON THE CLOCK)
   4 honor guards, 3 Snuggs, 3 paratroopers, 3 cubs · then EMPEROR BEAR (bar 233 × 2) at 3 s
2. **EN GARDE!** (HOLDS WHILE THE EMPEROR IS ON HIS HORSE)
   5 Snuggs, 3 cubs, 3 CAVALRYMEN

HARD · 22 bodies + the funnel

1. **EMPEROR BEAR TAKES THE FIELD** (ON THE CLOCK)
   4 honor guards, 3 Snuggs, 3 paratroopers, 3 cubs · then EMPEROR BEAR (bar 287 × 2) at 3 s
2. **EN GARDE!** (HOLDS WHILE THE EMPEROR IS ON HIS HORSE)
   6 Snuggs, 3 cubs, 3 CAVALRYMEN

MEGA · 22 bodies + the funnel

1. **EMPEROR BEAR TAKES THE FIELD** (ON THE CLOCK)
   4 honor guards, 3 Snuggs, 3 paratroopers, 3 cubs · then EMPEROR BEAR (bar 390 × 2) at 3 s
2. **EN GARDE!** (HOLDS WHILE THE EMPEROR IS ON HIS HORSE)
   6 Snuggs, 3 cubs, 3 CAVALRYMEN

The opening runs on the clock (the guards 1.6 s in, the Emperor at 3.0 s, the Snuggs 4.6 s, the paratroopers 5.4 s, the cubs 5.8 s). EN GARDE! waits for the keel: the Emperor's first bar at 0. On top of both: the battery at the ride's start, a Bluecoat detail every 9–14 s while he rides, a wave of Bluecoats and cannons every 8–15 s while he stands.

How this was read and checked

The live client v28.88 (index.html md5 1de0910be46cda954034c14925273a12) was opened headless on this Mac and read with the game's own functions: every table, every tier's wave sheet at the lab's own level, every man's HP and bar at every rank, the Emperor's and the cavalry's bars for 1 to 10 commanders, the price ladder. Then each hit was **measured on the real machines** in a BOSS LAB fight on every tier with the stock army worn (level 50 · 57 · 63 · 70), every plush born through the fields' own wave rows and the Emperor's own doors (the ride, the keel, the funnel, the battery): the cub's pounce on seven men (and 40 on the saber Sarge for his block), an honor guard beside a rifleman, a Snugg's and a paratrooper's glob to the KO, a Bluecoat's and the officer's dash-slash on men and on the saber Sarge (10 and 6 tries), the officer's whistle, the paratrooper's landing on eight men at 0–250 px (the reach 76 px from the landing point), a king's cannon on seven men and the jeep, the cavalry's charge through a line of eight, the Emperor's charge through the same line, his saddle swing on a rifleman, a saber Sarge, a pistol Sarge and King Rex, his battery, his first detail and a battalion, his three cuts on three men, the lunge from 300 px, the walk-in from 150 px, his cut on the saber Sarge (10 tries) and on each hero, MEN WITHOUT FEAR's wave and the final phase, the six heroes against the pounce, the glob, the lob and the hooves, King Rex at his last bar hit under a cavalryman's charge and under the Emperor's, every body as the tier bears it, the funnel's bodies at a party's price, the Emperor born in a party of 2, 3 and 5, and the keel through the game's own door (the snap, the second bar, EN GARDE!, the second pyramid, the EN GARDE! wave's release). Chrome 329/329 · WebKit 329/329 checks passed, zero page errors, and the two engines read every number the same.

Sources (the code's own names): RAID_INFO.fields, raidCfg's fields rows (tguard · bearnaparte · tsnugg · tpara · tcub · tcavalry; hpX kbPX · 0.6), FIELDS_HPX, RAID_CNT_MUL, BLB_STOCK_LV, blbStockDef, kbPriceX / kbBossX; FDEF rows; updateFoePounce → knock; tanShoot's pips → tanHit (DSM k 0.9, dsmX / dsmH, gritCap, the 4 s mend); updateFoePara (the THUMP → knock); guardDive; kbRoySeat / kbRoyFight / kbRoyTick / kbRoyForm, kbCut (KB_ROY_CUT_PIP), updateFoeGruffle (the officer); kbCanSeat / cannonTick / foeSplash (KB_COT_F, KB_COT_SCAT); dkStart / dkRideStep / kbTrample (KB_RAM_R, KB_TRAM_CD, KB_RAV_CD, KB_HOOVES); kbRideStart / kbRideStep / kbPickWay / kbGate1 / kbUnhorse / kbTick / kbFootTick / kbHuntFoot / kbWindGo / kbDashGo / kbDashTick / kbCut / kbAnswer / kbGate2 / kbMwfTick / kbBattery (KB_* cells); blbSheet('fields', tier, N), bhpX; bfHpAt, usBfHP, sgBfHP, heroBfHP, saberBlocks. Kit: design/boss-emperor-unit-breakdown-1031/ (kit/readBear.mjs · kit/proveBear.mjs · kit/build.py · kit/render.mjs · kit/export_dash.py; data/; proof/logs/). Zero game bytes.